CO1-AT2
ASSESMENT TOOL-2

ITA0329-MOBILE COMPUTING **13/10/2026**

K V CHARAN TEJA

192424167

Q1.

Given:

- Total bandwidth = 45 MHz = 45000 kHz

- Simplex channel bandwidth = 25 kHz

Formula:

Number of full-duplex channels = Total bandwidth / (2 × Channel bandwidth)

= 45000 / (2 × 25)

= 45000 / 50 = 900

Answer: 900 full-duplex voice channels.

Q2.

Given:

- Total channels = 560

- Cluster size = 7

Formula:

Channels per cell = Total channels / Cluster size

= 560 / 7 = 80

Answer: 80 channels.

Q3.

Given:

- Number of carriers = 100

- Time slots per carrier = 8

Formula:

Total voice channels = Number of carriers × Time slots

= 100 × 8 = 800

Answer: 800 simultaneous voice channels.

Q4.

Given:

- Base station power = 80 W

- Handset power = 4 W

Formula:

Power ratio = $10 \log_{10}(P_1 / P_2)$

= $10 \log_{10} (80 / 4)$

= $10 \log_{10} (20) = 13.01$ dB

Answer: 13.01 dB.

Q5.

Given:

- Carrier power = −60 dBm

- Interference power = −92 dBm

Formula:

C/I = Carrier power − Interference power

= −60 − (−92)

= −60 + 92 = 32 dB

Answer: 32 dB.

Q6.

Given:

- Total bandwidth = 42 MHz = 42000 kHz

- Channel bandwidth = 300 kHz

- Cluster size = 7

Formula:

Total channels = Total bandwidth / Channel bandwidth

Channels per cell = Total channels / Cluster size

Total channels = 42000 / 300 = 140

Channels per cell = 140 / 7 = 20

Answer: 20 channels.

Q7.

Given:

- Initial received power = 64 W

- Distance ratio = 2

- Path-loss exponent = 4

Formula:

$P_2 = P_1 / (\text{Distance ratio})^n$

$P_2 = 64 / 2^4$

$= 64 / 16 = 4$ W

Answer: 4 W.

Q8.

Given:

- Original radius = R

- New radius = 2R

- Path-loss exponent = 2

Formula:

$P_2 / P_1 = (R_2 / R_1)^n$

$= (2R / R)^2$

$= 2^2 = 4$

Answer: increased four times.

Q9.

Given:

- Number of cells = 4

- Channels per cell = 15

Formula:

Total channels = Number of cells × Channels per cell

$= 4 \times 15 = 60$

Answer: 60 channels.

Q10.

Given:

- Vehicle speed = 100 km/h

- Carrier frequency = 900 MHz

- Speed of light = 3×10^8 m/s

Formula:

Doppler shift = (Velocity × Frequency) / Speed of light

Velocity = 100 / 3.6 = 27.78 m/s

Doppler shift = $(27.78 \times 900 \times 10^6) / (3 \times 10^8)$

= 83.33 Hz

Answer: 83.33 Hz.

Q11.

Given:

- Frame duration = 5 ms

- Number of time slots = 8

Formula:

Slot duration = Frame duration / Number of slots

= 5 / 8 = 0.625 ms

Answer: 0.625 ms.

Q12.

Given:

- Total bandwidth = 24 MHz = 24000 kHz

- Carrier spacing = 200 kHz

Formula:

Number of carriers = Total bandwidth / Carrier spacing

= 24000 / 200 = 120

Answer: 120 carrier frequencies .

Q13.

Given:

- Transmitted power = 8 W = 8000 mW

- Path loss = 33 dB

Formula:

Transmitted power (dBm) = $10 \log_{10}$ (Power in mW)

Received power = Transmitted power − Path loss

Transmitted power = $10 \log_{10}(8000) = 39.03$ dBm

Received power = $39.03 - 33 = 6.03$ dBm

Answer: 6.03 dBm.

Q14.

Given:

- Frequency = 1900 MHz

- Distance = 2.5 km

Formula:

FSPL = $32.44 + 20 \log_{10}(f) + 20 \log_{10}(d)$

= $32.44 + 20 \log_{10}(1900) + 20 \log_{10}(2.5)$

= $32.44 + 65.58 + 7.96$

= 105.98 dB

Answer: 106 dB.

Q15.

Given:

- Number of cells = 6

- Calls per cell = 250

Formula:

Total capacity = Number of cells × Calls per cell

= $6 \times 250 = 1500$

Answer: 1500 simultaneous calls.

Q16.

Given:

- Call arrival rate = 4000 calls/second

- Average call duration = 3 minutes = 180 seconds

Formula:

Active calls = Call arrival rate × Average call duration

= 4000 × 180

= 720000

Answer: 720000 active calls.

Q17.

Given:

- Number of channels = 150

- Bandwidth per channel = 25 kHz

Formula:

Total bandwidth = Number of channels × Bandwidth per channel

= 150 × 25 = 3750 kHz

= 3.75 MHz

Answer: 3.75 MHz.

Q18.

Given:

- Number of frequency bands = 30

- Guard-band width = 40 kHz

Formula:

Number of guard bands = Number of bands − 1

Total guard-band bandwidth = Number of guard bands × Guard-band width

Number of guard bands = 30 − 1 = 29

Total bandwidth = 29 × 40 = 1160 kHz

Answer: 1160 kHz or 1.16 MHz.

Q19.

Given:

- Vehicle speed = 90 km/h

- Cell radius = 1.5 km

Formula:

Cell diameter = 2 × Radius

Handoffs per hour = Distance travelled / Cell diameter

Cell diameter = 2 × 1.5 = 3 km

Handoffs = 90 / 3 = 30

Answer: 30 handoffs occur per hour.

Q20.

Given:

- Regional centers = 8

- Base stations per center = 40

- Users per base station = 250

Formula:

Total capacity = Centers × Base stations × Users per base station

= 8 × 40 × 250

= 80000

Answer: 80000 users.

Q21.

Given:

- Total area = 400 km²

- Cell radius = 400 m = 0.4 km

Formula:

Cell area = R^2

Number of cells = Total area / Cell area

Cell area = $(0.4)^2 = 0.16$ km²

Number of cells = 400 / 0.16 = 2500

Answer: 2500 cells.

Q22.

Given:

- Total spectrum = 6 MHz = 6000 kHz

- Bandwidth per channel = 30 kHz

Formula:

Number of channels = Total bandwidth / Bandwidth per channel

= 6000 / 30 = 200

Answer: 200 FDMA channels .

Q23.

Given:

- Number of subscribers = 1200

- Calls per hour per subscriber = 4

- Average call duration = 2.5 minutes

Formula:

Traffic = (Subscribers × Calls per hour × Call duration) / 60

= (1200 × 4 × 2.5) / 60

= 12000 / 60 = 200 Erlangs

Answer: 200 Erlangs.

Q24.

Given:

- Total bandwidth = 40 MHz = 40000 kHz

- Bandwidth per user = 400 kHz

Formula:

Maximum users = Total bandwidth / Bandwidth per user

= 40000 / 400 = 100

Answer: 100 simultaneous users.

Q25.

Given:

- Number of base stations = 1200

- Users per base station = 600

Formula:

Total capacity = Number of base stations × Users per base station

= 1200 × 600 = 720000

Answer: 720000 simultaneous users.